BTBMS and finite Mahlo reflection

An independent, self-contained reconstruction

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Contents

Scope and proof architecture **2**

1. Syntax, order, and the fundamental operation **2**

2. Finite scan facts **4**

3. Closure, literal caps, and cofinality **6**

4. The finite source information needed later **14**

5. A hierarchy with explicit reflection certificates **16**

6. Configurations and their initial realizations **20**

7. Joint source packets **22**

8. Strict sources and semantic descent **25**

9. Deriving the full well-order **31**

10. Supplementary finite checks and their limits **32**

Scope and proof architecture

We work in ZFC and assume that for every natural number n there is an n-Mahlo cardinal. The intended conclusion is that the entire standard BTBMS order is a well-order. No well-foundedness of the syntax is used in any induction below.

There are three different kinds of induction. The finite calculus uses induction on finite syntax, column numbers, and finite generation witnesses. The semantic construction uses induction on the finite size of a target height. Inside one step of the latter induction, a counterexample is minimized by a pair of **actual ordinals**, not by the BTBMS order. Keeping these inductions separate is essential.

An occurrence is a node of a particular syntax tree. Different occurrences can have the same printed column number. Copying creates new occurrences; changing an address does not change the identity of a retained occurrence.

The proof is organized around four obligations:

1. The fundamental operations preserve the finite conditions needed by the scans.
2. A smaller standard expression is bounded by an approximant of any larger admissible expression.
3. Every seed has a finite ordinal configuration.
4. Every fundamental operation transports a configuration to one with smaller maximum.

The last section derives well-ordering from these four obligations. Bounded computational checks, described separately at the end, are not used in the mathematical arguments.

1. Syntax, order, and the fundamental operation

1.1 Finite syntax and visible columns

A forest is a finite list of columns. A column is a finite list of entries (p, H), where $p \in \mathbb{N}$ and H is a forest. All syntax trees are finite. The empty forest $\varnothing$ differs from the one-column forest $()$ whose only column is empty.

Entries are ordered first by p, then by H. Columns and forests are ordered lexicographically, with a proper prefix smaller. This defines a computable linear order by recursion on the two finite trees being compared. It does not assert that this linear order is well-founded.

The outer forest starts at column number 0. A height attached to column a starts at $a + 1$. Within each forest successive columns have successive numbers. When a height is entered, its containing columns remain visible; earlier columns of its own forest become visible as they are passed. On leaving a height its private columns cease to be visible. Thus a number on the visible stack denotes a unique occurrence.

We use the following conditions.

- **V:** an entry in column c has a visible parent $p < c$.
- **I:** every column of a nonempty height attached to a is nonempty, and its first parent is at least a.
- **H:** the entries of every column are nonincreasing.

For a host a, the relative key $h_a(H)$ replaces every parent in the whole height by

$$p \longmapsto \begin{cases} (0, p), & p < a, \\ (1, p - a), & p \geq a. \end{cases}$$

The same threshold a is used throughout this height, including all its descendants. The replacement is strictly increasing, so it preserves and reflects order.

Let $T_{r,d}$ add d to every parent at least r, at every depth. Let $A_{r,d,\ell,Z}$ do the same, except that it replaces the last column at depth ℓ on the last-column, last-entry route by the forest Z. The inserted forest is not shifted. At depth zero the formula is

$$A_{r,d,0,Z}(F) = T_{r,d}(F[:-1]) \frown Z.$$

At positive depth only the last entry of the last column is followed recursively; other heights are shifted without replacement. This specifies the operation even when Z is empty.

1.2 The two branches

The empty forest has no operation. If E ends in an empty outer column, $E[n]$ deletes that column for every $n \in \mathbb{N}$.

Otherwise follow last columns and last entries until the first empty height. The stopping column has the form

$$Q \frown [(r, \varnothing)].$$

Write t for its column number and ℓ for its depth. The forests along the route are $E_0 = E, \dots, E_\ell$, and

$$t = \sum_{i=0}^{\ell} |E_i| - 1.$$

The step is **special** when $\ell > 0$, Q is empty, and r is the direct owner of the stopping forest. If its containing column is

$$P \frown [(p, C \frown [[(a, \varnothing)]])],$$

replace it by

$$P \frown [(p, C)]^{n+\epsilon}, \qquad \epsilon = \begin{cases} 1, & P = \varnothing, \\ 0, & P \neq \varnothing. \end{cases}$$

All strict containers are retained and truncated after their active entries.

In all other cases the step is **ordinary**. Locate the visible occurrence r in a forest F at depth j and index i. Put $d = t - r > 0$. Scan the root heights, normalized by h_r, against the heights of Q, normalized by h_t. Both cursors start at zero. At keys x, y, advance the source cursor when $x \leq y$ and the query cursor when $x \geq y$. Stop when either list ends. Let U be the unconsumed root suffix.

Starting with $Z = \varnothing$, perform the following loop for $k = n, n-1, \dots, 1$.

- The new head consists of $T_{r,(k-1)d}(Q)$ followed by the entries of U. For an entry (p, H) of U, keep its outer parent p and replace its height by $T_{r,kd}(H)$.
- If $i < |F| - 1$, put $A_{r,kd,\ell-j,Z}(F[i+1:])$ after the head.
- If $i = |F| - 1$, require $U \neq \varnothing$. In its last restored height use $A_{r,kd,\ell-j-1,Z}$ instead of $T_{r,kd}$. The resulting head is the entire new forest.
- Assign the resulting forest to Z.

Finally set $E[n] = A_{0,0,\ell,Z}(E)$. In particular the ordinary operation at zero deletes the entire stopping column, not only its last entry.

1.3 Standard expressions and finite words

Let $L_0 = ()$. For $m > 0$, L_m has an empty outer column followed by a branch of m singleton entries. At column i the parent is $i - 1$; its height contains the next singleton, until the last height is empty. Define Std to be the expressions obtainable from these seeds by finitely many defined operations. A seed index and a finite list of parameters are a generation witness.

The compact word of a forest is defined at depth e as follows. An empty column contributes $(2e, 0, 0)$. The first entry (p, H) of a nonempty column contributes $(2e, 1, p)$ followed by the word of H. Every later entry contributes $(2e + 1, 0, p)$ followed by its height word. There are no closing symbols. Write $w(E)$ for the word and $\mu(E)$ for its length.

The word is injective: the depth coordinate identifies whether the next atom enters a child, continues a column, or starts a later column. The same observation proves that it preserves order. A child atom has larger depth coordinate than either kind of continuation outside that child; a continuation entry has larger depth coordinate than a new column. The empty-column and parent comparisons are exactly the other clauses of syntax order.

A **safe prefix** retains an initial segment of this word and closes its containers. Since the first entry includes its column marker, this operation cannot create an empty internal column.

2. Finite scan facts

For a finite word W in a linear order, let $\nu(W)$ be obtained by repeatedly deleting terminal letters strictly smaller than the next appended letter. Equal letters are retained. Thus $\nu(W)$ is nonincreasing.

For the relative height keys $K_0, \ldots, K_{m-1}$ of column c, put

$$B_c(-1) = \varnothing, \qquad B_c(i) = \nu(K_0, \ldots, K_i).$$

These boundaries strictly increase: an append either extends the reduced word or replaces its first removed letter by a larger letter.

Scan boundary lemma. The final source cursor against a reduced query W is the unique i such that

$$B_c(i - 1) \leq W < B_c(i),$$

or is m when $B_c(m - 1) \leq W$. Reducing a query does not change the final source cursor.

Proof. Deleting a source letter immediately followed by a larger letter does not change whether that prefix is exhausted against a nonincreasing query. Any query letter large enough to consume the larger one consumes the smaller one first; letters smaller than the larger one cannot exhaust the prefix. After reducing both nonincreasing lists, exhaustion is precisely lexicographic comparison, as is seen by cancelling equal initial letters and examining the first unequal pair or first ending. This proves the boundary formula. Deleting a query letter x immediately followed by a larger y does not change the source cursor after this two-letter block: every source letter consumed using x is also consumed using y. Repeating the deletion proves the last assertion. □

From now on, queries indexing g and its paths are reduced (nonincreasing) words. Let $g_W(c)$ be the parent of the first unconsumed entry, undefined if the scan is exhausted. A g_W-ancestor includes the starting vertex and the final vertex of its path. Condition V makes every nontrivial edge decrease the column number.

We add the condition

$$\mathbf{A}: \quad p_i \text{ is a } g_{B_c(i-1)}\text{-ancestor of } \begin{cases} c-1, & i=0, \\ p_{i-1}, & i>0. \end{cases}$$

All keys used in this definition are the actual relative keys of the explicit heights. Since every ancestor path decreases column numbers, A also implies $p_i \le p_{i-1}$ for $i > 0$. This parent monotonicity remains available for intermediate caps before H has been repaired.

Path nesting lemma. Under V and A, if $W \le Z$, each g_Z edge is a g_W path with the same endpoints.

Proof. Induct on the source column number, simultaneously in all scopes. If the Z cursor is i, the W cursor is some $j \le i$. If $j = i$ there is nothing to prove. Otherwise prepend the edge to p_j, and use the A paths from p_{k-1} to p_k, for $j < k \le i$. Their queries are $B_c(k-1) > W$. All their vertices are below c, so the induction hypothesis replaces their edges by g_W paths. Their concatenation has the required endpoint. □

An **owner return** consists of a height H attached to a and an entry (a, J) in a descendant column c. Require

$$\mathbf{S}: \qquad h_c(J) < h_a(H).$$

Let $\mathcal{L}$ be the class satisfying V, I, H, A, and S.

Barrier lemma. If the entry carrying H has index u, and the returning entry has index v, then

$$B_c(v) < B_a(u).$$

More generally, at any complete cut of any column, including the cut after its last entry, every read of an active containing entry u by the cut's ancestor path has query strictly below $B_a(u)$.

Proof. Use the following word observation. If $D = \nu(P \frown [K])$, W is nonincreasing, $W < D$, and $J < K$, then $\nu(W \frown [J]) < D$. All letters of D are at least K, so appending J cannot erase the first smaller decision or turn a proper prefix of D into a larger word.

Fix $D = B_a(u)$. Induct on a descendant column number and then on its entry index. A path witnessing a return to a either starts at a, or first reaches it by an edge $z \longrightarrow a$ with $a < z < c$. In the latter case its query is below the selected boundary in z, which is below D by the induction hypothesis. In the former case the query is empty or is the boundary of a preceding return. The new height key is below $h_a(H)$ by S, so the word observation gives the new boundary below D. For an arbitrary cut the identical start/last-edge argument applies; no following entry is needed. □

For use with changed heights, keys can be assigned independently of their explicit children. On the terminal containment route replace any selected last-entry key by a formal largest key $\top$, keeping its children. No entry query includes a changed last key.

Capping lemma. On inputs satisfying V and S, A is equivalent to A in each such independently capped system. Under V, I, A, S every operation is defined.

Proof. In the forward direction the barrier puts a query at or before the active entry. Raising that still-unconsumed key preserves its cursor and response. For the reverse direction induct on visible column number and entry index. On a capped path, a containing ancestor is reached from the start or through an edge at a smaller column. Those earlier entry assertions have already been uncapped. The proof of the barrier, using those assertions and S, puts this query below the actual active boundary. Its edge can therefore be uncapped too. This closes the induction.

Only the owner-root ordinary branch can be undefined. At its terminal return to r, with query Z, the barrier gives

$$Z < \nu(Z \frown [\varnothing]) < B_r(u)$$

for the active root entry u. The root scan therefore cannot be exhausted. □

We record exact trace facts, which are stronger than equality of the returned parent.

1. A common strictly increasing map on source and query keys preserves every cursor pair.
2. If a nonempty source is not exhausted, replacing its last key by $\top$ preserves every cursor pair.
3. If two sources agree before entry u, and their keys at u have identical comparison signs against every query key, then a run ending at or before u has exactly the same trace in both sources.

For (1), the comparison sign determines the next pair of cursors. For (2), once the last source entry is reached, every remaining query key must be strictly smaller, since otherwise that entry would be consumed. For (3), the two runs have the same signs up to the first possible move past u; the stipulated run never makes that move. These arguments also cover the empty query.

One way to verify the sign hypothesis in (3) is useful below. If two height words agree for more atoms than the length of a query height, their comparisons with that query height have the same strict sign. Their first disagreement is beyond the query's possible decision. Owner normalization changes neither atom positions nor word lengths.

3. Closure, literal caps, and cofinality

3.1 A profile used only for finite repairs

For an empty height set $\pi(\varnothing) = (-1, 0)$. For a nonempty height follow first-column, first-entry heights while they are nonempty. If there are d descents and the last first column begins with m maximal-parent entries, put $\pi(H) = (d, m)$. Those m heights are empty by H. Require

$$\mathbf{P}: \qquad \pi(J) < \pi(H)$$

for each owner return. Profiles are ordered lexicographically.

A literal cap at a column and cut k keeps the first k entries, appends $(p, \varnothing)$, and removes all later context. It is upward if it increases the next parent or appends at that column's end.

Profile and repair lemma. P implies S. Moreover, an upward cap of a V, I, H, P expression that satisfies V and I satisfies S. If it also has decorated A, finitely many same-parent moves turn it into an $\mathcal{L}$ cap with the same final parent and unchanged internal cap.

Proof. The first entry of the first column of a height must point to its direct owner. Thus the first-entry branch, after normalization, is a pure tower. Different profiles decide key comparison at the first differing tower depth or terminal multiplicity; hence P implies S.

For nonempty J, the greatest pointer-valid, I-legal upward literal cap is obtained by retaining this branch and appending one more maximal-parent empty entry to its terminal run. Call it $\Gamma(J)$. No earlier parent can be increased, since it is already maximal. A change away from this branch loses before the indicated append. Consequently

$$\pi(J) < \pi(H) \quad \Longrightarrow \quad h_c(\Gamma(J)) \le h_a(H).$$

In a changed owner return, the lower cap is at most $\Gamma(J)$ whereas its containing height has increased. Therefore

$$h_c(J') \le h_c(\Gamma(J)) \le h_a(H) < h_a(H').$$

Unchanged lower heights and newly empty returns satisfy S as well. Capping can now be undone.

Suppose a containing column violates H because its active last height C is larger than an earlier height with the same parent p. Move the entry carrying C from j to the first i with $H_i < C$, deleting entries $i, \dots, j-1$. The old column has $H_j \le H_i < C$. The elementary interval fact

$$C \text{ caps } X, \quad X \le Y < C \quad \Longrightarrow \quad C \text{ caps } Y \text{ at the same internal cut}$$

follows by inspecting the first differing atom. It shows that the moved subtree is the same literal cap.

In the decorated system the changed block is replaced by

$$P \frown [(p, K_i), \dots, (p, K_{j-1}), (p, \top)] \quad \longmapsto \quad P \frown [(p, \top)].$$

Every finite query has the same returned parent. The new outer entry's own ancestor test is the old test of entry i, since it has the same parent p and preceding prefix. Earlier retained tests are protected by the barrier. Tests inside the moved height see the same external responses. Profile safety permits uncapping again. The earliest choice of i repairs this column; an outer move carries the already repaired internal cap unchanged. There are finitely many containing columns. □

3.2 Structural closure

Closure theorem. The class $\mathcal{L}$ is closed under safe prefixes and all fundamental operations. The operations preserve P when present. Every standard expression lies in $\mathcal{L}$ and satisfies P. Every defined operation on a nonempty V, I expression strictly decreases it.

Proof: safe prefixes. We first isolate a word fact. If $X < Y$, a decrease of X first changes position α, and a decrease of Y first changes position $\beta > \alpha$, then the new X is still smaller. Also, changing Y only after position $|X|$ leaves $X < Y$ unchanged. Both statements follow at the first comparison decision and hold under a common order embedding.

In a shortened owner-return pair, the common global cut has position α in the lower height and $\delta + \alpha$ in its proper container, where $\delta > 0$. The word fact preserves S. If only the upper height is shortened, the complete lower occurrence ends before the cut, again giving the required length inequality. V, I, and H persist directly. Capping each affected active key before truncation preserves all retained A tests by the barrier; the established S permits uncapping afterward.

Profiles cannot increase under a decrease of height. If an upper profile changes at a safe cut, its first-entry branch reaches the cut. All preceding forests and columns on that branch are singletons, since otherwise the final cut would occur after the complete branch. The remaining height is a pure tower ending in a shorter maximal-parent run. A positive-depth return loses a tower level, and a depth-zero return has empty height. This proves P.

Proof: ordinary occurrence classes. Fix a positive ordinary step, put $d = t - r$, $s_j = r + jd$, and $\tau_j = T_{r,jd}$. Every output occurrence belongs to exactly one of the following classes:

1. original retained material, at phase zero;
2. a phase-j copy of a body column $r < x < t$, including all its closed side heights;
3. the query part of head s_j, copied with phase $j - 1$;

4. the root remainder in that head, whose outer parents are unchanged and whose heights have phase j.

This is an induction through the defining loop: the only non-pure copy follows its specified continuation hole; all its other children are pure shifts. The assertion therefore includes closed side heights whose printed numbers exceed t.

For a key normalized at $x \geq r$, define κ_j by adding jd to a false absolute field $(0, p)$ exactly when $p \geq r$, fixing the other fields. Then

$$h_{x+jd}(\tau_j H) = \kappa_j(h_x(H)), \qquad \kappa_i \kappa_j = \kappa_{i+j}.$$

These maps are order embeddings, commute with ν, and fix keys normalized at or below r.

First-parent edges form a preorder forest: A makes each new column attach to an ancestor of its predecessor, so after leaving the subtree of r the visible predecessor chain cannot return to it. Since the stopping edge at the complete query is to r, path nesting shows that every body column lies in that subtree. It is nonempty and has first parent at least r.

V now follows from matching shifts. The same observation proves I for copied body forests placed inside a new height. A nonempty query keeps its first-parent bound. For an empty internal query, the terminal parent r is strictly above the direct owner a, since equality would be special. The entire earlier root is restored and has first parent at least a; this verifies the remaining I case.

Query parents are at least r, while restored outer parents are below r. Thus their junction respects H. Elsewhere increasing parent maps preserve H. A changed continuation has its first difference at the old terminal: its last parent r is replaced by a restored parent below r, or its entry or column ends. Later repetitions are after this decision. Thus every changed height is smaller than its complete prototype, proving H at active boundaries and strict decrease of the entire expression.

Proof: all ordinary owner returns. A pure-copy upper height contains only copies of its old pairs. For a continuation upper height, distinguish its owner.

If the owner is a fixed $a < r$, the entire copied interval lies in that same old height. A parent equals a after shifting exactly when it did before. Every return therefore has an old origin in that height.

Otherwise the owner is $a + jd$ with $r \leq a < t$. Later body and restored-height phases $k \geq j + 1$ have parents either below r or at least $r + kd > a + jd$. Restored outer parents are below r. In a later query phase, equality with this owner requires

$$p + (k - 1)d = a + jd.$$

For $k \geq j + 2$ this would give $p < r$, contradicting that it was shifted. Only the first following query can return; its phase is j and its origin is the closed old query inside the same upper height. The list of returns is therefore exhaustive.

Transport an old pair by the common κ_j. For a fixed upper owner use the lower phase's map; it fixes both upper keys. When both heights change, the lower first-change position precedes the upper by its positive containment offset. When only the upper changes, the whole lower occurrence ends before that change. The word fact proves S in every listed case. This argument uses no output A assertion.

If an upper profile decreases while a new head remains, its first-entry branch reaches the old terminal. All intervening containers are singletons. The stopping column must consist of $m \geq 2$ maximal-parent empty entries, with $r = t - 1$ and $d = 1$; otherwise its leading profile would be unchanged, or the

step would be special. Its profile changes from (h, m) to $(h, m-1)$. Owners before $r-1$ have only their first chain return, which loses a level. The owner $r-1$ sees restored returns of depth zero or empty height. The owner r and its copies see only the empty entries of the next query. This proves P; continuation heights with no head are safe prefixes.

Proof: ordinary ancestor tests. Independently cap the continuation-dependent last keys in source and output. Let the assigned root keys be K_i, boundaries B_i, complete query Z, root cursor c, and $m = |Q|$. Capping preserves that actual cursor when the root is an owner. If c is not exhausted, then

$$B_{c-1} \le Z < B_c, \qquad \nu(Z \frown [K_c]) = B_c.$$

To verify the second equality, write $B_c = P \frown [K_c]$. The part removed from B_{c-1} lies below K_c. Any reduced word between these two boundaries has prefix P and then letters below K_c; appending K_c removes exactly that remainder. Iterating restores every later boundary.

At head j, put $Z_j = \kappa_{j-1} Z$. A query $V < Z_j$ stops inside its query prefix. If $V \ge Z_j$ and its old-root cursor is γ, its head cursor is

$$m + \gamma - c.$$

This includes exhaustion. The root keys are fixed by every κ_j.

For $V = \kappa_j W$, the high case $\kappa_1 W \ge Z$ gives the old root response at W. In the low case, the head's edge is a copy of the terminal edge at $\kappa_1 W < Z$. Path nesting supplies the terminal-to-root path. Its proper vertices lie between r and t, so their copies lead to the previous head. Repeat; the head index decreases each time. The process reaches a high exit or the original root, and then the fixed old prefix. Exhaustion is preserved as well.

It remains to check the starts and endpoints of each A test. Retained original entries have the same assigned fields. Body entries and entries in restored heights copy their source paths, using the head substitution at the root. An entry inside a query height can read its head only below that query entry's complete boundary, by the old barrier; it therefore uses the pure query part, after which the same substitution applies. Query-prefix entries copy the old terminal tests. At the first restored entry, the old terminal's last-entry test reaches r, and the next root edge supplies exactly its restored parent. At later restored entries the reset boundary is precisely the old root boundary, with both endpoints below r. Empty parts have no tests. These are the four occurrence classes already proved exhaustive. Decorated A follows, and S permits uncapping.

Proof: special steps and seeds. In a special step each replacement height C is smaller than $C \frown [[(a, \varnothing)]]$. The equal new entries and their separate height scopes give V, I, H and strict decrease. Returns cannot cross sibling scopes. The same containment-offset word argument gives S. A changed first-entry profile must have singleton containers, empty P, and empty C; it is a pure tower shortened by one level and ending in $n+1$ empty entries. Its returns lose one level or have empty height. Otherwise the profile is unchanged or the output is a safe prefix.

Cap the changed final entry and its active ancestors. Its old assigned suffix is $(p, \top)$; the new suffix consists of parent-p entries ending in $(p, \top)$. Every finite query returns the same parent. The first new entry uses the old test; later ones use the reflexive start p. Internal copies see the same external responses. Uncapping gives A. Finally the seeds have direct-owner parents, singleton columns, reflexive ancestor tests, and strictly shorter return profiles. Induction on generation proves the theorem. □

3.3 Kernels and transfer of successful caps

For an ordinary input H, let $F_H(X)$ be one copy of the expansion with its rightmost continuation hole filled by $T_{r,t-r}(X)$. Let C_H be the unchanged terminal context. Then

$$H[q] = C_H(F_H^q(\varnothing)), \qquad w(F_H(X)) = P \frown \Sigma(w(X)), \quad |P| > 0,$$

where Σ is a fixed letterwise order embedding (depth translation and parent shift).

Word criterion. $F_H(Y) > Y$ holds exactly when some iterate covers Y. Success is downward closed in Y. In the successful case every q with $q|P| > |w(Y)|$ gives a strict cover. A full cut of a one-column local target succeeds.

Proof. The iterates are the proper-prefix increasing words $P\Sigma(P)\cdots\Sigma^{q-1}(P)$. If $F(Y) < Y < X$, let i, j be the first decisions for these two inequalities. The first change from $F(Y)$ to $F(X)$ is at $|P| + j > j$, so the comparison at $\min(i, j)$ still gives $F(X) < X$. Failure is upward closed. An iterate covering Y therefore implies success. Conversely compare Y with the infinite concatenation of these blocks. If its first disagreement were a larger letter, $F(Y)$ would still read the corresponding earlier agreeing letter there and would be smaller than Y. Otherwise a sufficiently long iterate strictly exceeds Y. Equality $F(Y) = Y$ is impossible by length. At a full one-column cut the first output column extends the complete target column. □

We give an exact filling identity to avoid assumptions about rescanning. For a cap K ending at x with terminal prefix Q, root ρ and frozen actual cursor c, let $\mathrm{fill}(K, Y)$ replace its whole terminal column by the complete suffix Y. Let $\mathrm{rt}_{\rho,c}(D)$ be the root forest suffix with entries before c deleted from its first column; retain that column even if empty. Let $Q \diamond X$ prepend Q to the first column of nonempty X. Both operation branches give

$$F_K(Y) = Q \diamond T_{\rho,x-\rho}\, \mathrm{rt}_{\rho,c}(\mathrm{fill}(K, Y)). \tag{1}$$

In the owner-root case the insertion is inside its last restored height. The inserted Y is shifted exactly once. No scan of the filled object occurs.

When $Q \neq \varnothing$, delete $w([Q])$ from both sides of a comparison beginning with Q; when Q is empty delete nothing. Write this deletion as ∂_Q. Let e_0 be ordinary forest encoding; let e_1 encode the entries of the first column as continuation entries, omitting its marker when that column is empty. Then

$$\partial_Q w(Q \diamond X) = e_{\epsilon(Q)}(X), \qquad \epsilon(Q) = \begin{cases} 0, & Q = \varnothing, \\ 1, & Q \neq \varnothing. \end{cases} \tag{2}$$

Different prefixes must be deleted separately. Equal values of ϵ permit comparison of the resulting encodings.

Cap completion lemma. Let $B \in \mathcal{L}$ satisfy P, and suppose cofinality already holds with target B. An ordinary upward cap K of B with V, I and decorated A has a defined successful kernel at its complete old local suffix.

Proof. Profile safety and repair give an $\mathcal{L}$ cap K'. Cofinality and the word criterion give success there. Reverse one repair from entry i back to $j > i$ in column b. Let Y_i, Y_j be the complete old suffixes at the unchanged internal cap. The interval fact gives common retained prefixes; since $H_i \geq H_j$, it gives $Y_i \geq Y_j$.

If $b < \rho$, the root and hole are inside the unchanged moved subtree, so the kernels are identical and downward success applies. If $b = \rho$, all intervening old heights and the active cap have the same

strict sign against every query height: their common prefix contains that whole query height and its preceding entry atom. A query consuming entry i would consequently consume all intervening entries and the final active cap against that same key. This would exhaust the owner root. Hence the actual cursor is at most i, and exact trace fact (3) identifies both scans. If $b > \rho$, the root is unchanged, or only an owner's last active key changes. Both scans leave that key unconsumed, and exact trace fact (2) identifies them after capping.

In the latter two cases, completing Y_i and Y_j gives two safe truncations of B, the second properly extending the first. Root restriction and deletion of the common cursor prefix preserve this extension; in the case $b = \rho$, the bound $c \le i$ prevents deletion of its deciding part. Identity (1), with the same Q, x, ρ, c, gives

$$F_K(Y_j) > F_{K'}(Y_i) > Y_i \ge Y_j.$$

Reverse the finite list of repairs. □

Delayed-change lemma. Suppose $w(F_K(Y)) = P\Sigma(w(Y)) > w(Y)$. Put $N = \mu(Y)$ and $p = |P| \ge 1$. Delete a common prefix of length a from this comparison, obtaining $L > R$. If $R' < R$ and L' agrees with L at every position strictly before $p + N - 1 - a$, including existence of those letters, then $L' > R'$.

Indeed $p + N - 1 - a \ge N - a = |R|$. Thus L' preserves every possible decision against R, so $L' \ge R > R'$. The distinguished atom in applications is the inserted final atom of Y, at that displayed position. This verifies agreement lengths, not only parent-shift formulas.

3.4 Cofinality, with the cursor and cancellation checks

Cofinality theorem. If $B \in \mathrm{Std}$, $E \in \mathcal{L}$, and $B < E$, some n satisfies $B \le E[n] < E$. If E ends in a nonempty outer column, one can choose

$$1 \le n \le \mu(B) + 1, \qquad B < E[n].$$

Proof. An empty last column is the least possible column, so deletion already covers B. In the other case, $E[1] > B$ suffices, and $E[1] = B$ is handled by the next, strictly larger approximant. If $E[1] < B < E$ and the step is ordinary, $E[1]$ retains every atom before the final atom of E. Hence E is an upward literal cap of B. Kernel success at the complete local suffix then gives the claimed bound by the word criterion.

In a special step there is no I-legal height strictly between C and $C \frown [[(a, \varnothing)]]$, since the added column is the least permitted internal column. The target must therefore have height C at the first changed entry. H bounds each of its remaining entries by (p, C). More repetitions than their finite number give a strict cover within the displayed bound.

It remains to show kernel success for ordinary $\mathcal{L}$ caps of standard targets. Induct on a generation witness. A seed permits only full one-column upward cuts. A cap strictly before the end of a safe prefix of an already covered target also exceeds that target; a cap at its end is full. For a special output $P \frown [(p, C)]^m$, a cap inside the first copy of C exceeds the old target: its increase at an existing column precedes the appended least column. A cap in a later copy would exceed the preceding equal height and violate H. A nonfull cut between equal entries would need a parent larger than p and also violate H. Earlier and full containing cuts have already been handled.

Here is the ordinary induction step. Write $A_j = A[j]$ for an ordinary operation with period $d = t - r$, and let the new target be A_n, $n > 0$. A tested cap $H > A_n$ exceeding A is covered by the induction hypothesis. Thus assume $A_n < H \le A$. Nonfull cuts in unchanged material exceed A, and full

containing cuts are automatic. Every remaining cut is in a body column, a closed copied height, or a head. The occurrence classification in the closure proof exhausts these possibilities, including the final truncated copy.

Write $s_i = r + id$, $\tau_i = T_{r,id}$, and let ρ be the selected new root. We first justify projecting a new cut to an old cap. Undo the phase shift on its retained prefix. A proper current root $\rho > s_i$ projects to $\rho - id$, and a current head root $\rho = s_i$ projects to r. If the selected new root is the original r, stop its successful path at the current head s_i before projecting to r: from a current body, an edge either stays in the current phase or exits through that head or below r, and an exit below r cannot later reach r. The case $r < \rho < s_i$ will be handled directly and requires no old cap. For the remaining endpoints below r, fix a phase i, and express its reduced query as $V = \kappa_i W$. At head $h \le i$ put $W_h = \kappa_{i-h+1} W$, so $V = \kappa_{h-1} W_h$. If $W_h \ge Z$, the head's restored suffix has exactly the old root cursor at W: all root keys are fixed by κ. If $W_h < Z$, path nesting makes its old terminal path reach r before any smaller column, so its copy reaches head $h - 1$. Deleting these finitely many detours gives precisely the old continuation to any endpoint below r.

If the cut truncates a restored origin $u \ge c$, use the old root through u with that key capped. Its cursor at Z is still c. A high exit is its exact old response. A low exit has $W \le W_i < Z$; at such queries both the full and truncated root have cursor at most $c \le u$ and the same response. The later high exit has that response too, since both lists are κ-fixed. If the target is in a copied query height, its first head is the literal old truncated terminal field in phase $i - 1$, so that first edge projects directly; the same detour deletion applies afterward.

Cap all strict active ancestors while making this argument. Retained tests are protected by the old complete-cut barriers. Thus the projected old cap K has V, I and decorated A. It is upward by the same parent increase or append. It is ordinary: a singleton direct-owner column at cut zero is least and cannot be an upward cap. The cap completion lemma gives its definedness and success. No output ancestor assertion has been assumed in obtaining this cap.

Now compare the actual kernels. Let ρ be the new selected root and $s_i = r + id$.

- If $\rho > s_i$, root, scan and hole are pure phase-i copies, and

  $$F_H(\tau_i X) = \tau_i F_K(X).$$

  The complete new body suffix is $\tau_i W$ with $W \le Y$, where Y is its old complete suffix. Downward success proves the claim.
- If $\rho = s_i$, a high scan has actual head cursor $m + \gamma - c$, where γ is the old actual root cursor. To pass from assigned to actual keys, both containing-owner scans are first checked nonexhausting and then capped using exact trace fact (2). The entire query part is consumed, and the remaining root suffix, body and hole give the same conjugacy. A low scan restores a parent at least r, whereas the next target parent is below r or absent: a cap at s_i cannot increase a current-image parent at least s_i. It therefore wins at once.
- If $r < \rho < s_i$, a path reaching this previous-phase root must leave the current phase through a low head. A high exit would fall below r. In comparisons of the fixed $\kappa_i W$ with $\kappa_h Z$, $h < i$, moving false fields of the latter lie in $[s_h, s_{h+1})$, below s_i, whereas moving false fields of the former are at least s_i. Fixed false fields lie below r, and true fields follow all false fields. Thus the sign is independent of h, unless decided by identical fixed fields or a common ending, which also gives the same sign. Every reached positive head is low. At a proper previous root its next parent on the terminal-to-root path is at least r; the same holds at a previous head. Capping an unconsumed owner key does not change this actual response. It again beats the next target parent below r or extends its ending.

For completeness the fixed-root cases $\rho \le r$ require more than this response argument. If $\rho < r$, the old and new source fields agree except possibly at a containing owner's last active key; both are nonexhausting there. Cap them, use their κ_i-fixed keys, and apply exact trace facts (2) and (1). Their actual source cursors agree. For $\rho = r$ and a cut inside a restored origin u, the old projected key and the complete original key agree through more atoms than each terminal query height. The old owner scan ends at or before u. Exact trace fact (3) therefore identifies it with the full old-root scan. A possibly changed last active key in the new root is handled by fact (2). Direct body and query-height cuts need only this last-key argument. Let γ be the resulting common cursor.

The elementary shift identity is

$$T_{\rho,x+id-\rho} = \tau_i T_{\rho,x-\rho}. \tag{3}$$

Below ρ both sides fix a parent. At or above ρ, the first image under the right side is at least $x \ge r$, so the second shift adds id. Let θ_i undo τ_i on its image. We apply it only to remainders after their actual prefixes have been deleted.

For a closed side target let $Y_H = \tau_i Y$ and $Q_H = \tau_i Q_K$. Completing the old cut is a safe prefix of A ending before its terminal; completing the new cut extends this prefix into a later copy. Root restriction and deletion before γ preserve this proper extension, since an owner cursor retains the active origin. Both cancellations have the same $\epsilon(Q)$. Equations (1)–(3) give

$$\theta_i \partial_{Q_H} w(F_H(Y_H)) > \partial_{Q_K} w(F_K(Y)) > \partial_{Q_K} w(Y) = \theta_i \partial_{Q_H} w(Y_H).$$

This proves success in this case without the delayed-change lemma.

For a direct body target, let $Y_x(D)$ be the entire suffix beginning at the old column x. A cut retaining the original terminal entry would be full at a one-column local target; exclude it. Then

$$Y_K = Y_x(A), \quad Y_H = \tau_i Y_x(A_{n-i}), \quad Q_H = \tau_i Q_K,$$

$$\mathrm{fill}(K, Y_K) = A, \qquad \mathrm{fill}(H, Y_H) = A_n.$$

Indeed the whole suffix restores its forest, and every strict container is its last-entry, last-column container. Induction upward restores the whole expression. The loop places exactly $n - i$ remaining periods in the phase-i hole, including the case zero.

After separate cancellation and undoing the phase, the four remainders are

$$\begin{aligned} L_K &= e_\epsilon(T_{\rho,x-\rho}\,\mathrm{rt}_{\rho,\gamma}(A)), \quad & R_K &= \partial_{Q_K} w(Y_x(A)), \\ L_H &= e_\epsilon(T_{\rho,x-\rho}\,\mathrm{rt}_{\rho,\gamma}(A_n)), \quad & R_H &= \partial_{Q_K} w(Y_x(A_{n-i})). \end{aligned} \tag{4}$$

Let $w(F_K(Y_K)) = P\Sigma(w(Y_K))$, $p = |P|$, $N = \mu(Y_K)$ and $a = |w([Q_K])|$, with $a = 0$ for empty Q_K. The original terminal survives rooting: an earlier root retains its following forest and an owner cursor retains its active height. Its old position in L_K is $p + N - 1 - a \ge |R_K|$. Since $n > 0$, A and A_n agree before this terminal; (4) preserves that agreement. On the right the retained prefix excludes the terminal, so strict decrease gives $R_H < R_K$, also when $n - i = 0$. The delayed-change lemma gives $L_H > R_H$. Restore the phase and the new prefix.

It remains to examine cuts in head i. Put $m = |Q|$.

A cut $k < m$, or $k = m$ with closing parent $\rho > s_{i-1}$, is a phase-$(i-1)$ cap at the old terminal. Its root and kernel are pure copies. Undoing that phase leaves a target beginning with Q and then a parent below r, or ending after Q; it is smaller than the old terminal, so downward success applies.

At $k = m$, $\rho = s_{i-1}$, the cap retains $i-1$ complete periods and the unexpanded terminal. Its preceding active root height is therefore complete. The actual reset scan leaves exactly U, and the defining loop gives

$$H[q] = A[i-1+q] \qquad (q \geq 0).$$

Taking $q = n - i + 1$ reaches the target, hence yields kernel success. No smaller closing parent can be an upward cap: the actual complete-cut path first reaches s_{i-1} and then returns the first restored parent p_c, or is exhausted. The barrier permits capping without changing this response. Every smaller reachable endpoint is at most p_c, whereas upwardness requires one greater than p_c.

Finally take $k = m + v$, $1 \leq v \leq |U|$, and put $w = c + v$. A full owner-root head is a full one-column cut, already settled. In all remaining cases the retained restored heights are closed. Their actual keys are $K_c, \ldots, K_{w-1}$, and reset gives the query boundary B_{w-1}. H forces the new parent $\rho < r$. Cap the old root after w entries by this same parent. Its final test has exactly the same reduced query and prefix below r; cap completion applies. The actual selected-root cursors agree: identical closed sources use query reduction; containing owners are first capped using the exact nonexhaustion check, and then use query reduction. Only the final source cursors, not all raw query cursors, are identified here.

Write the old root column as R and its following forest as D. The old prefix is $Q_K = R[:w]$; the new one is the phase-$(i-1)$ query followed by the first v restored entries. Both are nonempty but can have different lengths. Their complete fillings are A, A_n. Removing these prefixes separately and undoing phase i gives

$$\begin{aligned} L_K &= e_1(T_{\rho,r-\rho}\,\mathrm{rt}_{\rho,\gamma}(A)), \quad & R_K &= e_1(\mathrm{rt}_{r,w}(A)), \\ L_H &= e_1(T_{\rho,r-\rho}\,\mathrm{rt}_{\rho,\gamma}(A_n)), \quad & R_H &= e_1(\mathrm{rt}_{r,w}(A_{n-i})). \end{aligned} \tag{5}$$

The mixed query prefix has been deleted before the inverse phase map is used. The original terminal survives: for an earlier root it lies in D, and for an owner root $w < |R|$ leaves its active height. With $a = |w([R[:w]])|$, its left position is again $p + N - 1 - a \geq |R_K|$. The left words agree before it, while the right word strictly decreases. The delayed-change lemma proves success. Empty U gives no cuts in this last group.

All cuts and selected-root positions have now been considered. This proves the ordinary induction step and hence the theorem. □

4. The finite source information needed later

A target is always a marked height B in one fixed original standard expression E. It need not itself be a standard expression in isolation. A current source is a marked height A' in an admissible context truncated after that height. We admit precisely the following records.

1. A **scan record** names an ordinary input H, either E or an admissible upward cap of E, its complete terminal query, its actual source index, and two closed retained heights A, B with $h_r(A) > h_t(B)$. The equation $g_Z(t) = r$ is asserted in H, not in E when these differ.
2. A **descendant record** retains this anchor, source entry, and pre-entry context, but replaces its source height by an admissible $A' \leq A$.
3. A **repair record** has source and target at the same printed host. Outside the source height, its context is obtained from the original target context by an explicit finite list of same-parent moves and removal of later continuation. Each move records the old and new address, common parent, retained later entry, and deleted closed blocks.

These are finite syntax records. They do not assume that any future target can be realized.

First-difference lemma. If a recorded source is strictly larger than B in relative order, a first parent-field difference or an additional source entry at an existing target column supplies an ordinary upward cap of E with V, I, A, S. Its closing parent is on the target's actual complete-cut chain.

Proof. In a scan record the common relative prefix shifts internal numbers by $t - r$ and fixes parents below r. A target false parent between r and t cannot occur in that common prefix: it would already be a relative difference. Thus common internal fields agree after the corresponding κ map; cap active keys while comparing the paths. A source endpoint at least r maps directly to its shifted endpoint. An exit below r sees the same fixed old fields. If the paths first reach r, t, the target internal-cut barrier puts its query below the boundary of its complete query member, hence below Z. Path nesting in H connects t to r at that query. The unchanged closed-prefix paths transfer it to E, and the fixed old continuation supplies the endpoint.

For a descendant record, first raise the assigned external active keys to their anchor values and restore unread later entries. These keys increase because the current height and truncated context decreased. Current cut barriers preserve the path until it reaches the source host or exits below it. Only there identify the old external fields; the preceding argument then applies. No preservation of the anchor's entire external query cursor is asserted.

For a repair record, each capped outer move replaces a parent-p block ending in $(p, \top)$ by $(p, \top)$. Every external query has the same returned parent. Common internal fields and these responses transfer the path directly. Profile safety uncaps the target.

The transferred parent increases the target parent or appends at that column. A changed first parent increases an I-legal one. A positive cut has at least two entries; a zero cut has parent strictly greater than the direct owner. Hence the cap is ordinary. Retained tests use the old cut barriers, and the new closing test uses the transferred complete-cut path. Profile safety gives S. □

Source reduction lemma. These records have the following closure properties.

- Safe prefixes and actual operations confined to the marked source preserve its record type and admissibility.
- Repair of a first-difference cap takes place outside its marked internal height. If the final old target at its new address is B_{good}, then $B_{\text{good}} \geq B$. Every positive output of the repaired cap strictly above E has marked height $B' > B_{\text{good}} \geq B$ and a repair record for the original B.
- If a proper child forest of the target ends first, a safe source truncation followed by one actual operation gives a smaller source height still strictly above B.

Proof. The first claim follows from structural closure; no prefix or external stack before the marked entry changes.

For the second, the internal cap is a reindexed safe prefix of the current source, so H already holds internally. Repairs can move only its containing entry or a strict outer container. At one move let the old heights be $X_j \leq X_i < C$, and let the marked old heights at their common internal address be $B^{(q)}, B^{(q+1)}$. The interval fact identifies the prefixes before that address. If $B^{(q)} > B^{(q+1)}$, their first differing atom would give $X_j > X_i$ before either outer continuation is read. This also holds for proper-prefix comparisons, since an internal continuation atom has greater depth than an outer continuation atom. Thus

$$B = B^{(0)} \leq \cdots \leq B^{(m)} = B_{\text{good}}. \tag{6}$$

The same internal cap is unchanged throughout. Every terminal-query height J is a complete earlier

child of this cap already present in the original B. If N counts internal ordinary and entry vertices, then

$$N(J) + 2 \leq N(B), \tag{7}$$

because its containing column and entry contribute two further vertices. This bound uses B, not B_{good}.

The actual operation replaces the terminal hole inside the marked height and retains that original marked entry. Earlier syntax agrees with E and later outer continuation is truncated. A resulting $B' < B_{\mathrm{good}}$ would put the whole output below E; equality would give a safe prefix of E. An output above E must therefore have the stated strict inequality. Cofinality supplies a positive such output.

For the third claim, truncate the source through the first entry of its extra column and clear that entry's height. Let its parent be p and its direct child owner a. I gives $p \geq a$. If $p > a$, the operation has empty query and an earlier root in that child forest; a positive output keeps the whole target child forest and adds a nonempty forest. If $p = a$, the special operation repeats the entry whose shortened height is exactly that target child forest. H bounds all following target entries by this entry, so sufficiently many repetitions decide a strict comparison. The direct child owner is internal to the marked source because this is a proper child difference. Thus the marked outer source entry remains, and closure and strict decrease give

$$h_t(B) < h_r(A'') < h_r(A').$$

If the whole target height ends first, it is already a proper source prefix and no such operation is needed. These cases exhaust first differences of finite syntax. □

5. A hierarchy with explicit reflection certificates

5.1 Construction and canonical hulls

All coordinates below are limit ordinals greater than ω; abbreviate this by $\mathrm{Lim}(a)$. We repeatedly use the elementary fact

$$v < u,\ \mathrm{Lim}(u) \quad \Longrightarrow \quad v + \omega \leq u. \tag{8}$$

Use Kuratowski pairs and right-associated tuples. Fix the following finite family $\mathcal{B}$ of operations; $r[a] = \{z : \langle a, z\rangle \in r\}$ denotes a fibre:

$$\{x, y\}, \quad x \setminus y, \quad x \times y, \quad \bigcup x, \quad \mathrm{dom}(x), \quad x^{-1},$$
$$\{\langle u, v\rangle \in x^2 : u \in v\}, \quad \{\langle u, u\rangle : u \in x\},$$
$$\{\langle u, z, v\rangle : z \in x, \langle u, v\rangle \in y\}, \quad \{\langle u, v, z\rangle : z \in x, \langle u, v\rangle \in y\},$$
$$\{x[z] : z \in y\}, \quad x[y], \quad \langle x, y\rangle, \quad \langle x, y, z\rangle, \quad \{x, \langle y, z\rangle\}.$$

Each has a bounded graph: quantify over the input sets, the proposed value and their members, and check its defining equality in both directions.

For transitive X, let $X^+ = X \cup \{X\}$ and define

$$S(X) = X^+ \cup \{G(\vec{x}) : G \in \mathcal{B},\ \vec{x} \in (X^+)^{\mathrm{arity}(G)}\}.$$

This is transitive. Members of set-operation outputs lie in X^+. Members of relation outputs are the listed pair or triple outputs. Members of a fibre-family output are listed fibre outputs. The supports

of pairs and triples are the listed pair and wrapper outputs. These cover every generator. The graph of S is bounded by its input and asserted output.

Define

$$J_0 = \varnothing, \quad J_{\xi+1} = S(J_\xi), \quad J_\lambda = \bigcup_{\xi<\lambda} J_\xi \quad (\lambda \text{ limit}).$$

At a nonzero limit, J_λ is transitive and $\mathcal{B}$-closed.

We spell out the coding facts needed for reflection. In a transitive closed level containing a whole set T, the displayed operations construct the truth table of each fixed formula over $(T, \in)$. Products give assignment sets; the restricted membership relation and diagonal give atomic tables; difference gives complements; insertion places variables in their required tuple positions; converse and domain give projections. An existential quantifier is a projection of its subformula table, and a universal quantifier is complement followed by projection and complement. Induction on the formula constructs all tables by finitely many operations.

There is one bounded verifier for a supplied record of such tables, even when its finite formula code varies. The record includes the parse tree, powers of T, and every subformula table. Check each product and table equality in both directions, bound every assignment quantifier by the recorded assignment set, and require the distinguished assignment to be in the root table. Induction on the parse tree proves correctness and uniqueness. For a bounded formula and transitive T containing its parameters, its truth over T is absolute to the universe.

The construction of $S(T)$ inside a larger closed level does not require collecting generator values by an unproved closure principle. For transitive T, finite truth-table operations form the incidence relations for unordered pairs and fibres. Their fibre families give

$$P_T = \{\{x, y\} : x, y \in T\}, \qquad F_T = \{r[a] : r, a \in T\}.$$

The same pair construction works on $U = T \cup (T \times T)$. Every member of a generator value on arguments from T belongs to

$$B_T = T \cup (T \times T) \cup (T \times (T \times T)) \cup F_T \cup P_U.$$

For each generator form the bounded relation

$$\{\langle \vec{x}, z\rangle : \vec{x} \in T^k,\ z \in B_T,\ z \in G(\vec{x})\}.$$

Taking its fibres over T^k collects exactly the required values. Tuple bounds needed here are obtained by finitely many products and unions. Applying this to T^+ constructs $S(T)$ inside every transitive closed level containing T.

For every nonzero limit λ,

$$\operatorname{Ord} \cap J_\lambda = \operatorname{rk}(J_\lambda) = \lambda, \qquad |J_\xi| \le \max(\aleph_0, |\xi|). \tag{9}$$

For size use finite arities and continuity. For height, the first ω stages give the hereditarily finite sets. Induct in blocks of ω stages. Each generator increases rank by a finite amount, giving the upper bound in the next block. The ordinal part of the previous whole limit level is a fixed-formula subset of that level, so the table construction places it, and then every finite successor, in the next block. This gives the lower bound. Limits of blocks use unions.

History and hull lemma. The map $\xi \mapsto J_\xi$ is globally single-valued Σ_1. Its history through ξ belongs to $J_{\xi+\omega}$. There are compatible globally single-valued partial Σ_1 Skolem functions on all J_λ

with $\mathrm{Lim}(\lambda)$. Their hulls are Σ_1 elementary, monotone, continuous on increasing unions, and have size at most $\max(\aleph_0, |X|)$ when generated by X.

Proof. A history is a function on $\xi + 1$ satisfying the displayed successor and union clauses. Its check is bounded, including both directions of every union equality. External recursion proves uniqueness.

Decorate each stage with a well-order. At a successor place old elements first, then the old stage itself, then genuinely new generator values ordered by their least production code (generator index followed by its argument tuple in the old order). At limits take unions. Fixed finite lexicographic products of well-orders are well-orders, by successive minimization of coordinates. The bounded production checks and external induction give unique end-extending decorated histories.

At a block boundary $\eta = \omega\alpha$, every shorter history already belongs to J_η. Thus the history below η and the order at η are fixed-formula relations over J_η: recover a stage or comparison by a checked shorter history. Their entries belong to J_η by finite coding closure. The truth-table construction places these relations in $J_{\eta+\omega}$. Append the endpoint fields η, J_η and any prescribed finite continuation there. This block induction proves the budget for both histories. Existence of a boundedly checked history is their Σ_1 definition.

For any x, its predecessors in the decorated global order form a set I_x, obtained by restricting the order of an earlier stage containing x. End-extension makes it independent of that stage. If $x \in J_\lambda$, the relevant history and restriction belong to J_λ by (8).

For a code i of $\exists y\, \vartheta(y, \vec{p})$ with bounded ϑ, let $V(i, \vec{p}, q)$ check a witness y, a transitive bound T, and the supplied true tables for ϑ. This is one bounded predicate. Any witness in a closed limit level has such a record in that level, by choosing an earlier transitive stage containing its parameters and performing the finite construction. Choose the least valid record using

$$V(i, \vec{p}, q) \ \wedge\ \forall q' \in I_q\ \neg V(i, \vec{p}, q').$$

Supplying the genuine decorated history makes this a Σ_1 definition. All predecessors of a record belong to the same limit level as that record, so the definitions agree on every common domain of definition. Closing under these functions gives bounded elementarity by induction on bounded formulas and then Σ_1 elementarity by the witness property. Countably many finitary functions closed in ω steps give the size bound; finite dependence gives monotonicity and continuity. □

Condensation lemma. If $H \prec_{\Sigma_1} J_C$ and $\mathrm{Lim}(C)$, its transitive collapse is J_δ for a limit $\omega < \delta \leq C$. If $u, J_u \in H$, the collapse sends J_u to $J_{\pi(u)}$.

Proof. A symmetric-difference witness reflected into H makes its membership relation extensional. It is well-founded as a subrelation of membership. Its collapse N is closed under $\mathcal{B}$, since the generators have bounded graphs and their existence reflects. Elementarity puts every finite ordinal and ω in H; successor closure shows that $\delta = \mathrm{Ord} \cap N$ is a limit greater than ω.

For $\nu < \delta$, its preimage is an ordinal because ordinalhood is bounded. A genuine history through this preimage is available in J_C by (8). Reflect it into H, collapse it, and use bounded absoluteness and history uniqueness. Its endpoint is J_ν. Hence $J_\delta \subseteq N$. Conversely every preimage of $x \in N$ belongs to some earlier stage, with history in J_C. Reflect this existence statement and collapse it; then $x \in J_\nu$ for some $\nu < \delta$. This proves equality. The inverse collapse increases ordinals, giving $\delta \leq C$. The same history argument proves the final clause. □

5.2 Control edges and complete certificates

For $a < b < C$ with all three limit coordinates, define

$$R(C, b, a) \iff \mathrm{Hull}_{\Sigma_1}^{J_C}(J_a \cup \{b\}) \cap J_b = J_a \quad \text{and} \quad J_b \prec_{\Sigma_1} J_C. \tag{10}$$

An edge implies $J_a \prec J_b$: the hull contains the whole J_b by its canonical history, and for any whole set T in a Σ_1 elementary hull, relativization of an arbitrary formula to T is bounded. Witness reflection gives $H \cap T \prec T$. Apply this to $T = J_b$.

The edge remains true when its environment C is replaced by a limit D with $b < D \leq C$. Canonical function compatibility gives the smaller hull as a subhull, while it still contains all generators. The trace remains exactly J_a, and Σ_1 upward absoluteness gives the intermediate elementarity. It also remains true for $C \leq D$ when $J_C \prec_{\Sigma_1} J_D$, since all canonical evaluations on old arguments then have the same domains and values.

The control hull collapses to J_δ. It fixes J_a pointwise, and its intersection with b is a, so it sends b to a. In particular $a < \delta \leq C$.

A finite diagram uses limit coordinates, strict inequalities, edges (10), Σ_1 elementary pairs, and explicit shared variables. Its **complete certificate** includes its assignment, genuine histories, and checked true tables for every asserted relation. Auxiliary sets and labels are allowed only with separately established budgets.

Certificate lemma. There is a bounded checker for complete certificates. Existence of a complete certificate is consequently Σ_1. A finite diagram of maximum v has a certificate in $J_{v+\omega}$. For additional set parameters and auxiliary records, the same conclusion holds when they have that budget and the parameters used inside the top-level tables belong to J_v.

Proof. Uniform Σ_1 satisfaction in a genuine limit level L is

$$\mathrm{Sat}_1^L(i, \vec{p}) \iff \exists q \in L \; V(i, \vec{p}, q).$$

Canonical evaluations can likewise be expressed with their witnesses bounded by L. Membership in a hull is existence in L of a finite evaluation trace whose lines are generators or canonical evaluations on earlier lines. Finite packing in L proves that this formula is exactly hull membership.

Thus two fixed first-order formulas express, for genuine levels $A \subseteq B \subseteq L$, both $A \prec_{\Sigma_1} B$ and the trace equation

$$\forall x \in B \, [x \in \mathrm{Hull}_{\Sigma_1}^L(A \cup \{b\}) \leftrightarrow x \in A].$$

Their tables over the whole $M = J_v$ can be constructed by finitely many generators. For a coordinate $c < v$, J_c and its history are in M by (8). The top level J_v is a tag for the ambient universe M, not an element of M: quantifiers over it become unrestricted quantifiers in the formula over M. Externally check $v = \mathrm{Ord} \cap M$ and all top-pin identifications. Neither v nor M is incorrectly used as an element parameter of that formula.

The whole M, its history, all these tables and finite coordinate records belong to $J_{v+\omega}$. A bounded checker verifies histories, both directions of table equalities, and truth at each distinguished assignment. Bounded auxiliary checks can be performed on the supplied sets externally. For a variable bounded-formula code choose a transitive stage containing its finitely many parameters inside the closed $J_{v+\omega}$ and use the same table verifier. Finite packing gives the claimed certificate. Conversely a passing record has genuine levels by history uniqueness and true assertions by table correctness. □

Capture and transport. If $J_a \prec_{\Sigma_1} J_b$, fixed parameters lie in J_a, and a complete existence problem has a witness in J_b, it has a complete witness in J_a. Every coordinate in the captured record is below a by transitivity and (9).

Under $R(C, b, a)$, a complete problem with fixed parameters in J_a, distinguished parameter b, and witness in J_C transports to that problem with distinguished parameter a. Reflect the whole record into the control hull and collapse it. Bounded checks remain true, certified whole levels collapse to their genuine levels, and shared variables remain shared. The resulting witness lies in $J_\delta \subseteq J_C$ and can be captured into J_b, since its new parameters belong to J_b.

This transports one existentially chosen joint record. It does not assert a pointwise transformation of an arbitrary supplied assignment into an arbitrary captured assignment.

6. Configurations and their initial realizations

6.1 The finite graph of an expression

Read syntax depth first. Give each column occurrence an ordinary role A_c, and each entry e a slot U_e, placed immediately before its height. On the visible stack each column p has ordinary representative $O(p) = A_p$ and current representative $P(p)$. Entering an entry at column p replaces $P(p)$ by that entry's slot; exiting restores the previous representative.

An entry (p, H) in column c contributes

$$R(U_e, A_c, O(p)), \qquad R(U_e, A_c, P(p)), \tag{11}$$

with duplicates omitted. Mark every role with Lim, order them strictly by reading order, and require $J_{A_c} \prec_{\Sigma_1} J_{A_{c+1}}$ for consecutive columns in each individual forest. There is no elementary pair between distinct sibling height forests. Call this graph $\Gamma(E)$. A configuration is an ordinal assignment satisfying its graph. The empty expression has the empty configuration.

For an ordinary step use the roles R, T, N of root ordinary, stopping ordinary, and final control slot. Let Q now denote the roles strictly between T and N. From the interval (R, T) remove every consumed root entry slot and its whole height; call the remainder K, and set

$$D = \{x : x < R\} \cup \{R\} \cup K, \qquad P = \{x : x < T\}.$$

Every retained edge reads only D; every query edge reads only $D \cup \{T\} \cup Q$. Indeed a consumed root entry is closed (an owner's active entry cannot be consumed), and its private roles have left the stack before later surviving entries. Removing sibling heights creates no ordinary pair between their forests.

Occurrence-map identity. Begin with the old restriction to P and $f_0(x) = x$ on D. At stage $m+1$ append a head h, a fresh query copy, and a fresh retained copy. The query map is

$$q_{m+1}(T) = h, \quad q_{m+1}|_Q = \text{fresh query copy}, \quad q_{m+1}|_D = f_m.$$

The retained map is

$$f_{m+1}(R) = h, \quad f_{m+1}|_K = \text{fresh retained copy}, \quad f_{m+1}(x) = x \ (x < R).$$

Use the query map for query edges, internal query pairs and the pair entering T, if present; use the retained map for retained edges and pairs. After n stages this is exactly $\Gamma(E[n])$.

Proof. Query syntax is shifted in the previous phase, retained syntax in the new phase. Stack entry and exit commute with these occurrence substitutions. The continuation is a rightmost forest hole and is inserted once without an additional shift. These facts identify every edge by induction on the loop. Ordinary pairs inside copied forests use their indicated map. A root-to-next-column pair uses the retained map. If T has a predecessor a in its own forest, the new entering pair is $(f_m(a), h)$, including an active-root hole; otherwise there is no such pair. Removing whole sibling heights introduces no other pair. The initial restriction is precisely parameter zero. □

There is a different persistent map

$$\iota_m : P \hookrightarrow \Gamma(E[m]) \tag{12}$$

onto the original pre-stop occurrences. Appending a stage never changes these names. Generally ι_m and f_m differ; old labels must use ι_m, never silently substitute f_m.

6.2 Mahlo initialization

Let M_0 be the inaccessible cardinals and let

$$M_{n+1} = \{\kappa \in M_0 : M_n \cap \kappa \text{ is stationary in } \kappa\}.$$

The axiom is $\forall n \in \omega\ \exists \kappa \in M_n$.

If λ is an uncountable cardinal and $u > \lambda$ is a limit, then

$$J_\lambda \prec_{\Sigma_1} J_u. \tag{13}$$

To prove this, put the parameters of a true Σ_1 statement in a transitive J_ξ with $\xi < \lambda$. Its hull in J_u has size at most $\max(\aleph_0, |\xi|) < \lambda$. Condensation collapses it to J_η with $|\eta| < \lambda$, hence $\eta < \lambda$. The collapse fixes J_ξ pointwise by membership induction. Its image witness therefore has the original parameters and belongs to J_λ. Bounded absoluteness gives the reverse implication.

If λ is uncountable regular, there is a club $C_{\lambda,u} \subseteq \lambda$ such that

$$a \in C_{\lambda,u} \quad \Longrightarrow \quad R(u, \lambda, a). \tag{14}$$

For $r < \lambda$ let $H_r = \mathrm{Hull}^{J_u}_{\Sigma_1}(J_r \cup \{\lambda\})$. Its size is below λ, so regularity gives $f(r) < \lambda$, $f(r) > r$, with $H_r \cap J_\lambda \subseteq J_{f(r)}$. Take the limit ordinals $a > \omega$ closed under f. They are unbounded: iterate countably often above any prescribed bound, at each step exceeding $f(s)$ for every s below the current bound; regularity keeps the supremum below λ. They are closed because every s below an increasing supremum lies below an earlier member. At such an a, continuity of levels and finitarity of the hull give

$$\mathrm{Hull}^{J_u}_{\Sigma_1}(J_a \cup \{\lambda\}) = \bigcup_{r<a} H_r, \qquad \mathrm{Hull}^{J_u}_{\Sigma_1}(J_a \cup \{\lambda\}) \cap J_\lambda = J_a.$$

Together with (13), this proves (14).

Seed lemma. Every L_m has a configuration.

For $m \geq 1$ its reading order is $A_0, A_1, U_1, \ldots, A_m, U_m$. The required edges are

$$R(U_1, A_1, A_0), \qquad R(U_i, A_i, A_{i-1}),\ R(U_i, A_i, U_{i-1}) \quad (2 \leq i \leq m).$$

The only additional outer pair is implied by the first edge. Choose $a_m \in M_{m-1}$ and $u_m = a_m + \omega$. Descending from $i = m$ to 2, stationarity selects

$$a_{i-1} \in C_{a_i,u_i} \cap M_{i-2},$$

and unboundedness selects $u_{i-1} \in C_{a_i,u_i}$ above a_{i-1}. Finally choose $a_0 \in C_{a_1,u_1}$. Assign $A_i = a_i$, $U_i = u_i$. The displayed inequalities and (14) verify every edge and reading-order inequality. For $m = 0$ assign its sole ordinary role $\omega \cdot 2$. □

7. Joint source packets

7.1 Names, frames, and certificates

Every role has an immutable occurrence name and an instance tag. A chart injects the roles of one finite syntax graph into such names. Different charts overlap only on explicitly shared names. Each chart retains its own internal orders; distinct private names in different charts need not be ordered and may have equal ordinal values.

For an entry e of F, let $P(F, e)$ be the full graph restriction to roles strictly before its slot. This need not be admissible syntax: it can end at an ordinary role before that column's first entry.

A height's frame consists of its physical host ρ and the ordinary and currently visible representatives of **every** externally visible column below it. Unused representatives are retained, since a later cap can read a larger parent. Include selectors for the two representatives of the outer entry's parent. All frame coordinates are at most the host and precede the outer owner.

For fixed target syntax B, a positive attached block consists of a new owner s, the complete internal graph of B interpreted at this frame, and both target head edges

$$R(s, \rho, p_o), \qquad R(s, \rho, p_v). \tag{15}$$

A free block omits only these heads. A floor L requires all new coordinates, including s, to exceed L. The syntax, host convention and frame selectors determine an independent bounded checker

$$\mathrm{Cert}_B(z; \rho, \vec{c}, L).$$

It is defined before and independently of any source assertion. For a limit u (and frame below u), a positive realization with maximum $v < u$ has a complete certificate in $J_{v+\omega} \subseteq J_u$. Conversely a certificate in J_u has every recorded ordinal below u. Define the bounded failure predicate, with a supplied whole set U, by

$$\mathrm{No}_B(U; \rho, \vec{c}, L) \iff \forall z \in U \ \neg\, \mathrm{Cert}_B(z; \rho, \vec{c}, L). \tag{16}$$

A **source packet** for a finite record of Section 4 consists of:

1. the complete target-prefix chart $P(E, b)$ for the original target B;
2. the complete source context graph, with source prefix $P(F, a)$, physical host ρ, and outer owner u;
3. a finite auxiliary diagram with already supplied certificates and bounded labels;
4. one assignment on all these names, agreeing on every specified overlap;
5. both target heads (15) at the same u, in addition to the source's own heads;
6. one complete joint certificate $w_{\mathrm{low}} \in J_u$ for both prefixes and the auxiliary diagram, with every component assignment checked against the corresponding projection of that one assignment.

The two physical host roles share ρ. In a cross-host record the true external ordinary and visible roles below the source printed host agree. In a same-host repair record the entire external frame agrees

under the recorded surviving-role map. The owner u, its private height and the whole J_u are not part of the low certificate. In particular no certificate containing u is asserted to belong to J_u.

To transport a joint record, quantify one assignment $a : I \to \mathrm{Ord}$ on its finite name set. For every component chart ξ, require its certificate's assignment to be exactly $a \circ \xi$. Require every label's host, owner and frame arguments to be those same named projections. These are bounded equalities. Capture or control collapse applied to the whole checked record preserves them. Separate realizations would not suffice.

7.2 Fresh target variables and first-difference caps

Factorization lemma. Given a packet and $L < u$, there is $L_* < u$, $L_* \geq L$, such that every positive target certificate in J_u above L_* is compatible with all unchanged low resources.

Proof. By continuity choose $\beta < u$ with $w_{\mathrm{low}} \in J_\beta$. Every component is a finite nested field of that record, hence belongs to J_β by transitivity. Choose $L_* < u$ above β, the host, and every low coordinate, and at least L.

Give the new target fresh private names, sharing only its prescribed frame. The old and new assignments agree on that entire intersection. Every old atom and label retains its old arguments; every new atom is checked by the target certificate. All needed cross-prefix order inequalities follow from the raised floor. There is no elementary pair across a height attachment or between sibling heights. This verifies all atoms of their union.

If $s > L_*$ is the new owner, the old record belongs to $J_\beta \subseteq J_s$. Its genuine stage history can be supplied separately in $J_{\beta+\omega} \subseteq J_s$. Finite packing gives the joint certificate in J_u. Thus failure of the resource-preserving conclusion implies $\mathrm{No}_B(J_u; \rho, \vec{c}, L_*)$. No parameter has been dropped from a predicate that still depends on it. □

Joint cap lemma. At a first parent-field or existing-column-end difference $h_r(A) > h_t(B)$, the finite cap from Section 4 has a joint realization with the old low resources. Its marked outer owner is the original u. If W, C are its stopping ordinary and closing slot values, then $u < W < C$. Its whole pre-stop graph, old resources, and genuine J_u with history have a joint certificate in J_W. Any already true bounded label with checking data in $J_{u+\omega}$ can be included, in particular (16) at $U = J_u$.

Proof. Equal initial relative syntax identifies the two internal stacks. A false parent is the same external parent below r; its two representatives agree by the packet. A true parent denotes the corresponding host or an already matched internal column. Entering matched entries installs matched visible representatives, and exiting restores them. Thus every internal edge and ordinary pair agrees. The last source entry supplies both head edges of the closing target entry; the finite first-difference lemma has separately established its actual ancestor legality.

Use the original target prefix unchanged, use the source owner u for its marked slot, and use the corresponding source roles for the common internal prefix and closing entry. This is one specified union of existing assignments, agreeing on their overlap. Its new internal coordinates exceed u; no elementary pair crosses the attachment. The pre-stop finite graph has maximum $v < W$, so its certificate lies in $J_{v+\omega} \subseteq J_W$. Old low records lie in J_u, while the whole J_u and its history lie in $J_{u+\omega} \subseteq J_W$. A true label with that budget can be packed and checked there. Its budget proves availability of its checking data, not its truth; truth is a premise. □

Joint repair lemma. Outer repairs preserve this joint realization, the marked host and owner, its full frame, and all retained names and values. Deleted blocks remain in the auxiliary diagram. The

repaired prefix before the marked owner has an injective chart into the old target prefix, with its original low budget.

Proof. A move retains the later active entry and its slot s and removes only earlier same-parent closed entries. Its heads have the same physical host and parent representatives. Entry into its height therefore installs the same slot, so all internal edges see the same frame. Removed private roles have already left the stack. Deletion does not reorder surviving roles or introduce a pair between sibling forests. Restriction of the old assignment realizes the new main graph; the old assignment still realizes the removed auxiliary blocks. Every deleted coordinate precedes $s \leq u$, so its graph certificate has budget below u. More directly, the deleted blocks already belong to the old target prefix certificate, which can be kept unchanged.

Iterating records an explicit surviving-role injection. Moving an address does not identify the retained later slot with the old earlier slot. All low records can be repacked in J_u, placed in one J_β with $\beta < u$, and the floor raised above β. Raising the floor preserves a true negative predicate (16). □

These lemmas also verify packet admission after the reductions of Section 4. A source operation confined to its marked height retains its prefix and frame. A repaired positive cap output above E has height $B' > B$ by (6) and the explicit same-host move record. The source prefix is the surviving restriction of the original target prefix, while the full latter remains auxiliary. Every subsidiary query height is a retained occurrence of E at the cap's **current** address, and is also the same unchanged proper child of the original B for purposes of (7). The current occurrence and the original failure target are different fields of the record.

7.3 Complete packets at a scan

The following precise admission fact will be used in the semantic induction. Let H be a standard input or an admissible cap of an original standard E. Let P^- be its whole graph before its stopping ordinary role. Supply:

- a transported chart of all of P^- and old auxiliary data, with one certificate in J_ρ;
- a new stopping role of value ρ and a complete positive query, jointly realizing its edges and its entering ordinary pair when present;
- a separate source bank through closed root entries $A_0, \ldots, A_{m-1}$, with original values, whose root is that same new role ρ and whose whole pre-root chart is shared with the transported chart.

All other bank names are fresh. In particular its root is not the transported reference occurrence of the old root, which lies below ρ.

At raw scan state (i, j), retain the already fitted positive query entries of indices less than j with fresh names, complete syntax and unchanged full frame. Let L_j be their entire target-prefix maximum, with $L_0 = \rho$, and let C_j be the corresponding original positive-prefix maximum. If

$$L_j < u_i, \qquad L_j \leq C_j, \tag{17}$$

and the target heads at source owner u_i hold, then the next source-greater comparison has a complete source packet.

To check this, the source prefix is the full bank restriction before u_i, including its pre-root chart and all earlier closed entries. The target prefix is the entire transported P^-, the new ordinary role, and all fitted earlier query entries. Its domain is exactly $P(E, b_j)$ at the current cap address: every

query entry precedes the new closing entry, so all its earlier syntax is retained from E. This includes consumed root blocks and the entering ordinary pair, which cannot be omitted.

The two assignments share the root value ρ and the whole fixed pre-root chart, and agree there; other private names are disjoint. All auxiliary labels retain their old arguments. There are no unlisted cross-chart pairs or orders. Every coordinate in this union is below u_i by (17) and bank reading order. If v is their finite maximum, its graph certificate lies in $J_{v+\omega} \subseteq J_{u_i}$. The old record in J_ρ has the same budget. Reconstruct component certificates for the present prefixes and check their projections against the union assignment. This gives one joint low certificate. Transient certificates from previous fitting steps are not kept as old resources unless their budgets are separately established.

A source-only advance adds one closed bank block to the prefix, below the next owner. A target advance adds a complete freshly named positive block above L_j. Its internal edges use only its frame and new roles; old edges read no new role. Exiting its height restores the stack. There is no pair to another sibling height. Thus the prefix assertion persists whenever the updated bounds (17) hold. The original positive query remains a separate candidate throughout, even if some of its numerical values are reused. This proves admission by induction on the actual finite scan trace, without assuming any new target exists.

8. Strict sources and semantic descent

8.1 The induction assertions

Let $N(B)$ count the ordinary and slot roles internal to B, excluding its outer owner. Define S_n as follows:

> For every original standard target occurrence B with $N(B) = n$, every admitted finite strict-source record $h_r(A) > h_t(B)$, and every source packet realizing it at physical host ρ and owner u with both target heads at u, each floor $L < u$ admits a complete positive target certificate in J_u above L, compatible with all unchanged low resources.

All frames, charts, assignments and low budgets are quantified. Only the original target expression is required to be standard. Old bounded labels cannot read fresh target-private names, and their joint certificates must already lie in J_u.

We prove, for each n,

$$S_{<n} \quad \Longrightarrow \quad W_{<n} \quad \Longrightarrow \quad D_{<n} \quad \Longrightarrow \quad S_n. \tag{18}$$

Here $W_{<n}$ fits a complete query, all of whose child heights have size below n, below its final unconsumed closed source owner. Its endpoint hypothesis is that the actual raw scan does not exhaust the source bank. The assertion $D_{<n}$ realizes ordinary successors below their old stopping ordinary value, transporting the specified original roles and labels. Its inputs are standard expressions and admissible caps of original standard expressions, with the query children and closed scan sources retained at their current original occurrences. It also applies to any admissible ordinary input with empty query; this case needs no source assertion or standard anchor.

For a nonempty attached block with owner h, every private coordinate z satisfies

$$J_h \prec_{\Sigma_1} J_z. \tag{19}$$

The first child column's first parent is the direct owner by V and I, so its visible-parent edge starts this path. Consecutive ordinary pairs traverse a forest. Each entry edge supplies elementarity from

its ordinary value to its slot. Its child's first visible edge continues the path. Induction through the finite tree proves (19).

We also need a two-candidate fact. Two realizations of the same free-block problem above the same floor admit a realization whose owner and maximum are no larger than the corresponding owners and maxima of either candidate. Choose X of smaller maximum, breaking ties by smaller owner. If its owner is already smaller, use X. Otherwise

$$h_Y < h_X \le c_X < c_Y.$$

These inequalities force Y to be nonempty, so its owner path reaches c_Y. The certificate of X lies in $J_{c_X+\omega} \subseteq J_{c_Y}$. The frame and floor lie in J_{h_Y}, and (19) gives $J_{h_Y} \prec_{\Sigma_1} J_{c_Y}$. Capture the free existence problem there. Its new maximum is below h_Y, proving all bounds.

The base S_0 is direct. A strict source for an empty target is nonempty. Its first ordinary value b satisfies $J_u \prec_{\Sigma_1} J_b$. After raising the floor by factorization, u itself witnesses in J_b the existence of a limit $s > \max(\omega, L_*)$. Capture gives such an $s < u$. Restrict the two target heads from environment u to s. Its empty positive block has a certificate in $J_{s+\omega} \subseteq J_u$, and factorization supplies compatibility.

8.2 Fitting a query: $S_{<n} \Rightarrow W_{<n}$

Use the complete bank and prefix data of Section 7.3. Write source keys $x_i = h_r(A_i)$ and query keys $y_j = h_t(J_j)$, source owners u_i, and source maxima M_i. Reading order gives

$$u_i \le M_i < u_{i+1}.$$

Keep the original positive query in a separate candidate chart. Maintain (17). It holds initially because $L_0 = C_0 = \rho < u_0$. All old reference and auxiliary data retain their certificate in J_ρ.

If $x_i < y_j$, advance only the source cursor. The fitted target prefix is unchanged, and $L_j < u_i \le M_i < u_{i+1}$ gives the next bound. The source advance has a next owner because the stipulated final cursor is unconsumed. No strict-source call is used.

If $x_i > y_j$, let h, c_0 be the owner and maximum of the original positive block for J_j. Its owner is above $C_j \ge L_j$, so it is a candidate above the current floor. If $c_0 < u_i$, reuse its values on fresh names. Otherwise establish both target heads at u_i before calling S. When $h \ge u_i$, restrict their environments. When $h < u_i \le c_0$, the block is nonempty; (19) and intermediate Σ_1 elementarity give $J_h \prec_{\Sigma_1} J_{u_i}$, permitting their environments to increase. Section 7.3 now supplies the entire joint packet, including both prefixes. Apply $S_{N(J_j)}$, whose index is below n. It gives a complete block above L_j with maximum below $u_i \le c_0$, preserving all old data. Advance only the target cursor. Both inequalities (17) persist.

If $x_i = y_j$, equality of the full relative syntax identifies the same free problem wherever an external role is read. False parents read the shared genuine prefix below r; true parents read corresponding internal roles. Add unused full-frame fields to the source candidate without imposing new conditions. Apply the two-candidate fact to the source block and original positive target block. Its owner is no greater than the positive owner's, so restricting the positive heads supplies both heads. Use fresh target names and advance both cursors. Its maximum is at most $M_i < u_{i+1}$ and at most the original positive maximum C_{j+1}. This also covers empty equal heights.

The prefix and joint-certificate checks of Section 7.3 apply at every step. Each step decreases $(m-i)+(q-j)$, so the scan terminates. Nonexhaustion of the bank forces exhaustion of the query; its final prefix maximum is below u_c. The finite complete query and old data can be certified there.

For an empty query, $c = 0$ and the conclusion is simply $\rho < u_0$, without any packet call. This proves $W_{<n}$.

8.3 Ordinary descent: $W_{<n} \Rightarrow D_{<n}$

Fix an allowed ordinary input and a supplied configuration. Use the role sets of Section 6.1 and write

$$p = \alpha_R < W = \alpha_T < C = \alpha_N, \qquad R(C, W, p). \tag{20}$$

The final control edge supplies (20). Let an old bounded label Λ refer to named roles of P, auxiliary resources, and genuine whole levels at those role coordinates. It may not refer to a future query or copied role. Assume a single joint pre-stop certificate for this graph, resources and label in J_W. The outside control value C is not included in a certificate claimed to belong to J_C.

Fix the pre-root roles at their original values. Any additional finite tuple already in J_p may also be fixed. All other data are existential variables with explicit sharing and budget conditions. Choose $\eta < p$ above all fixed pre-root coordinates and stages containing the additional fixed parameters. Only nonfixed **main** and newly introduced reference coordinates will be required to exceed η; auxiliary charts keep their own conditions.

Let G_m be the actual graph after m copy stages. Maintain a realization below W with latest root p, latest retained copy at exactly the original values $\alpha|K$, and nonfixed main coordinates above η. Its graph, old resources and label have one joint certificate in J_W. At zero this is the original restriction to P.

At each stage add a fresh **reference chart of the whole original** P. It shares D with the latest main copy through f_m and assigns those roles the original values. Its consumed blocks $P \setminus D$ have fresh names and original values. This assignment agrees on every shared role. There is no order between independent private charts. Its certificate can be packed with the maintained graph and old label into one $w_{\mathrm{old}} \in J_W$.

Append temporarily the head W and the original positive query, reading the latest $f_m(D)$, including the entering ordinary pair. Its atoms hold by the occurrence-map identity and original values. Its finite maximum is below C, so it has a certificate in J_C. The whole J_W, its history and w_{old} also have that budget by (8). Pack them in one checked temporary state.

Express existence of this state with distinguished parameter $t = W$. Fixed parameters are exactly the pre-root data, chosen low tuple, η and finite codes. Include all chart equalities, the main and new reference inequalities $\eta < x < t$, all old auxiliary conditions and the genuine-level budget

$$w_{\mathrm{old}} \in J_t. \tag{21}$$

Apply control transport under (20). The distinguished head becomes p, and all preceding nonfixed main and new reference coordinates become less than p, still above η. The whole old-state certificate now belongs to the genuine J_p. Capture the transported assertion into J_W using $J_W \prec_{\Sigma_1} J_C$. Its new fixed parameters, including p, belong to J_W. We now have a positive query above p with maximum below W, and the transported main graph, reference chart and labels certified in J_p.

The query must next fit before the restored K. If K is empty it already fits. If its first role v is ordinary, ordinary pairs along the remaining forest and entry edges along the terminal route give $J_v \prec_{\Sigma_1} J_W$. If its first role is an active owner, the owner path gives the same relation. In either case capture the entire transported existence problem into J_v. Its parameters belong there, and the condition (21) still certifies the old state in J_p.

The remaining case is that $v = u_c$ is a closed root owner at the scan's final cursor. Restrict the original root bank to entries $0, \dots, c$. This contains the entire actual scan trace and its unconsumed endpoint; all these entries are closed. Give it original values and a complete pre-root chart. Share only that fixed chart and the new head p with the transported data; other bank names are fresh. The transported reference occurrence of R remains a different role below p.

The transported reference chart contains the entire old P, including the consumed blocks, and the positive query has its entering ordinary pair. Its joint old certificate is in J_p by (21). Thus every hypothesis of Section 7.3 and $W_{<n}$ is present. The original standard or current cap occurrences supply the finite scan records. Apply $W_{<n}$ to fit the complete query below v, changing only its fresh private variables. If the query is empty this is immediate and does not require a standard anchor, which proves the advertised extension to arbitrary admissible empty-query inputs.

Now restore K at its original values, with fresh names and new root p. Its edges read only the fixed pre-root data, the new root, and this new retained copy. Query edges still read the transported preceding copy, exactly as q_{m+1} specifies. Its entering pair and all retained pairs have been checked. The union is therefore exactly G_{m+1}.

Old labels continue to read $\iota_{m+1}(P)$, not the restored latest copy. Their certificate remains in J_p. Every actual graph coordinate is below W, so the new graph and old data have a joint certificate in J_W. This proves the stage invariant and hence every finite positive operation. Zero uses restriction.

In particular, for an original marked entry e strictly containing the stopping column, its host, owner and frame in the output are exactly the projections of the transported old record onto

$$\iota_k(h), \qquad \iota_k(U_e), \qquad \iota_k \circ j, \tag{22}$$

where j is its complete original frame map. The syntactic operation retains all these original occurrences; new copies have fresh names. Shared-variable checks make these simultaneously the arguments of all transported labels and budgets. If the input root value is at most the old marked host value, its new host is below p, since it is a nonfixed old main role. If its prefix and failure data are before the root and belong to J_p, they can instead all be fixed. This proves $D_{<n}$ including its precise marker and label clauses.

8.4 Special descent and its limited label clause

Suppose a special step shortens $C_0 \frown [[(a, \varnothing)]]$ at entry e. Let s be its owner and W the removed last ordinary value. The final visible-parent edge gives $J_s \prec_{\Sigma_1} J_W$.

The owner s, its completed height prefix C_0, and its two outer heads form a positive block of maximum below W, certified in J_W. Its external frame is below s. Choose a floor $L < s$ above the retained main prefix, any fixed low coordinates, and stages containing the finitely many fixed auxiliary records. The displayed original block is a witness above L in J_W. Capture its positive existence assertion into J_s, quantifying its owner. This gives one fresh copy with all new coordinates between L and s and a certificate in J_s.

For each further copy raise L above the last maximum and a stage containing its certificate. Continuity at the limit s permits this finitely often. Make exactly $n + \epsilon$ copies. Their only shared data are their full external frame; the stack is restored on exit and no elementary pair joins sibling height forests. Their union is precisely the special output. Every coordinate is below $s < W$. Zero copies leave the prefix.

A joint label can be kept fixed here only if its certificate and parameters already belong to J_s and it reads no newly inserted private role. In particular this does **not** permit fixing the whole J_s as a parameter of capture into J_s. Strict containing entries, their names and their frames remain unchanged.

8.5 Strict sources: $D_{<n} \Rightarrow S_n$

Assume $S_{<n}$ and thus the established $W_{<n}, D_{<n}$. Suppose S_n fails. Choose a failure with least ordinal pair (ρ, u) in lexicographic order. This is legitimate in ZFC without a syntax well-order: one failure bounds a nonempty set of possible hosts; Separation yields a least host, and one failure there bounds a nonempty set of owners with a least member.

The chosen failure includes its original E, B, finite source record, full joint packet and floor. Factorization supplies a stage β and raised floor L_* such that

$$w_{\mathrm{low}} \in J_\beta, \qquad \beta < L_* < u, \qquad \mathrm{No}_B(J_u; \rho, \vec{c}, L_*) \text{ is true.} \tag{23}$$

The whole J_u is a parameter of the last assertion externally, not an element of its own low certificate. Follow the first difference of the complete relative keys of the strict source and target.

Case 1: the whole target ends first. Its corresponding complete source prefix, with owner u and the supplied target heads, realizes the positive target above L_*. If M is its maximum and z the first omitted source vertex, $M < z$ and its certificate belongs to J_z. The owner path gives $J_u \prec_{\Sigma_1} J_z$. Capture positive existence there, quantifying the owner and fixing only frame, floor, low records and their prefix projections, all in J_u. The resulting positive target certificate lies in J_u, contradicting (23).

Case 2: a proper child forest ends first. Apply the source reduction lemma: truncate through the extra ordinary column and its first entry, clear that height, and take a suitable actual operation. Its root ordinary value is strictly above u, since its direct child owner is internal to the marked source. If ordinary, its query is empty and the unrestricted empty-query case of $D_{<n}$ applies. If special, use Section 8.4 and enough repetitions to exceed the finite target run. Both give an admissible smaller source $A'' > B$ with its same descendant or repair record.

All original prefix, frame, floor and low records can be fixed: they lie in J_u, below the operation's internal control. In the special case its local owner s is strictly above u; the retained marked owner and both head relations, as well as the required low records, have certificates in J_s. This meets precisely the limited special label clause. The whole negative assertion at J_u is kept externally and is not a parameter of the next capture into J_u.

Let W be the extra ordinary value from the truncation. The new complete source, including its own and target heads, has a joint certificate in J_W, and the old owner path gives $J_u \prec_{\Sigma_1} J_W$. Capture this source existence problem into J_u, quantifying a new owner $w > L_*$ and fixing the full low record, its projections and stage β. Its witness has $w < u$. Since $\beta < L_* < w$, the same low record lies in J_w. Its host and frame are unchanged, both sets of heads use this same new owner, and its finite source record is unchanged. It is an admitted packet at (ρ, w). Since $J_w \subseteq J_u$, (23) excludes every positive target certificate there at the same floor. This is a smaller failure, a contradiction.

Case 3: a parent differs or an existing column ends first. The finite first-difference lemma gives an ordinary upward cap of the original E. The joint cap lemma realizes it with the original marked owner u and values $u < W < C$ at its internal stop and closing slot. It also supplies a joint pre-stop certificate for the already true negative assertion in (23): its finite parameters lie in J_u, and its whole-level parameter, history and checking data lie in $J_{u+\omega} \subseteq J_W$.

Perform all required outer repairs. They preserve the marked owner, full frame and internal cap. Keep deleted blocks as low resources. Repack the complete low diagram in J_u, choose one new containing stage $\beta < u$, and raise L_* if needed. A larger floor preserves the negative assertion. Let H be the repaired cap, at current marked address b_m, while the original target address remains b_0.

The internal cap is unchanged, so (6) gives $B_{\mathrm{good}} \geq B$, and every terminal-query height J satisfies $N(J) + 2 \leq n$ by (7). Each such J is also a retained occurrence of the original standard E at the cap's current address. These give exactly the smaller-size and original-standard hypotheses for $D_{<n}$ and its subsidiary scan calls. Cofinality gives a positive k with $H[k] > E$. Its height B' at the original marked entry satisfies $B' > B_{\mathrm{good}} \geq B$.

We now verify explicitly that semantic descent produces an admitted smaller failure for the **same** original E, B.

Let e_* be the retained marked entry. Let V_S be its repaired source-prefix role set, $\ell : V_S \to P_H^-$ its inclusion in the actual pre-stop graph, and $\chi : V_S \to P(E, b_0)$ its surviving-role chart. The input joint record identifies these projections. Their domains are not silently replaced by the old unrepaired prefix. The finite source context F_k is the actual output $H[k]$ through the complete height B' at the original e_*. Its prefix and external repair list are exactly the recorded ones.

Use one assignment a on all named pre-stop and auxiliary roles. Let h name the marked host, a_* its slot, and j_0 its full frame map. The source and negative label share literally the fields

$$a_h, \quad a_{a_*}, \quad (a_{j_0(v)})_v. \tag{24}$$

Supply a whole set U and its history with $U = J_{a_{a_*}}$, and check

$$\mathrm{No}_B(U; a_h, (a_{j_0(v)})_v, L_*), \qquad w_{\mathrm{low}} \in J_\beta, \quad \beta < L_* < a_{a_*}. \tag{25}$$

Every component certificate in w_{low} has its assignment checked against the appropriate projection of a. This whole higher record belongs to J_W: the negative label is already true, its whole-level data have budget $J_{u+\omega}$, and the pre-stop graph and low resources have the established budgets. Thus the labelled premise of $D_{<n}$ is actually met.

The whole external frame survives every repair. A deleted block is a closed sibling, whose private roles have left the visible stack. Active ancestor slots and visible ordinary columns are retained. Entry into the retained later slot installs the same representative and exit restores the same saved one. Induction through the containers proves this also for unused external representatives.

Let p be the actual root value of H.

If $p \leq \rho$, use the moving-parameter clause of $D_{<n}$. Only its allowed pre-root data and finite codes are fixed; (24)–(25), including the whole level and failure predicate, travel in the joint check. Let the output assignment be a' and put

$$\rho' = a'_h, \qquad u' = a'_{a_*}, \qquad \vec{c}' = (a'_{j_0(v)})_v.$$

The marker identity (22) makes these precisely the actual host, owner and full frame of F_k, as well as the arguments of the transported label. Genuine histories identify its whole set as $J_{u'}$, so

$$\mathrm{No}_B(J_{u'}; \rho', \vec{c}', L'_*) \tag{26}$$

holds with the original finite target code B.

Here are all admission checks. Restriction of the actual output graph realizes F_k and its marked height $B' > B$. The target prefix retains domain $P(E, b_0)$, including deleted blocks as auxiliary names. Its source prefix is the actual $\iota_k \circ \ell$. If γ is the main output assignment, the transported sharing check is

$$\gamma(\iota_k(\ell(v))) = a'_{\chi(v)} \qquad (v \in V_S).$$

Thus the prefixes agree on every required shared role, with their stated domains. The whole-frame argument verifies all external identifications. The target heads use $u', \rho', \vec{c}'$ from (24); the source heads belong to the actual output graph. Finally (25) transports to

$$w'_{\mathrm{low}} \in J_{\beta'}, \qquad \beta' < L'_* < u'.$$

Its component projection checks are still true, so it is the required single joint low certificate in $J_{u'}$. All packet clauses hold, and (26) says it fails at that floor. The marked host is an old main role at or after the root; a positive stage puts it below p, and subsequent stages keep it there. Hence $\rho' < p \leq \rho$, a smaller host-owner pair.

If $p > \rho$, the root lies strictly inside the marked height: every visible external parent has ordinary value at most ρ. Thus $u < p$. All marked prefixes, frame and low records are before the root and lie in J_u; the whole J_u, its history and negative checking data lie in $J_{u+\omega} \subseteq J_p$. They may therefore be fixed in $D_{<n}$. The result realizes F_k with the original host ρ, owner u, full frame and both sets of heads, and has maximum below W. Together with its fixed low data it has a certificate in J_W.

Use the old internal path $J_u \prec_{\Sigma_1} J_W$ for a second, distinct capture. Its fixed parameters are exactly

$$(E, b_0, B),\ F_k,\ \chi,\ \rho,\ \vec{c},\ L_*,\ w_{\mathrm{low}},\ \beta.$$

They all belong to J_u. Quantify a new owner $w > L_*$, the private coordinates of B', and their complete certificate. Require the full source graph with its fixed prefix projections and **both** source and target heads at the same w. The whole J_u and the negative assertion are absent from this existence problem. The source just constructed at owner u witnesses it in J_W.

Capture gives $L_* < w < u$. The finite syntax record and target code remain unchanged. The host and full frame were fixed, the two head clauses were checked at the new owner, and $w_{\mathrm{low}} \in J_\beta \subseteq J_w$ supplies all joint low budgets. This is an admitted packet at (ρ, w). Externally restrict (23) from J_u to J_w; it is still a failure at the same floor. Again the pair is smaller.

Both root positions contradict minimality. Together with Cases 1 and 2, these exhaust every finite first difference. Thus S_n holds. Induction on n proves (18) at all finite sizes. The only minimal-counterexample argument used actual ordinal pairs after constructing a new packet in exactly the same admitted class. □

9. Deriving the full well-order

Theorem. ZFC plus the existence of an n-Mahlo cardinal for each $n \in \omega$ proves that $(\mathrm{Std}, <)$ is a well-order.

Proof. Every seed has a configuration by Section 6.2. The completed induction gives ordinary descent for any standard input: choose n larger than the sizes of all its finitely many query heights. Section 8.4 gives special descent. Deletion restricts a configuration and removes its largest role. Consequently, if a nonempty standard A has configuration α, every nonempty $A[k]$ has a configuration γ such that

$$\max \gamma < \max \alpha. \tag{27}$$

In the ordinary case the new maximum is below the old stopping ordinary; in the special case it is below the removed last ordinary; each is below the old maximum. Induction on a finite generation witness therefore configures every standard expression.

Suppose an infinite descending standard sequence $E_0 > E_1 > \cdots$ existed. None of its terms is empty. Take a configuration of $A_0 = E_0$ with maximum θ. All subsequent choices can be made in the set

$$X = \omega \times \mathrm{Std} \times (\theta + 1)^{<\omega}.$$

A state (i, A, α) is eligible when A is standard, $A \geq E_i$, and α is its correctly indexed configuration. From such a state, $E_{i+1} < A$. Cofinality supplies k with $A[k] \geq E_{i+1}$. The output is nonempty and standard. By (27) it has a configuration with strictly smaller maximum, giving another eligible state in X.

Choose a well-order of this set by Choice and recursively select its least eligible successor satisfying these requirements. Ordinary recursion on ω produces an infinite strictly descending sequence of ordinal maxima, impossible. Finally, a nonempty subset of the linear order Std without a least member would produce a descending sequence by the same set-sized choice recursion. Hence every nonempty subset has a least member. This proves the full well-order, not only an initial segment. □

An explicit ordinal interpretation can now be recorded without choosing configurations canonically. Set $\sigma(\varnothing) = 0$ and, for nonempty standard E, define

$$\sigma(E) = \min\{\max \alpha + 1 : \alpha \text{ realizes } \Gamma(E)\}.$$

The class in braces is nonempty; choose one member to bound the minimum search and use Separation. Applying (27) to a minimizing configuration gives $\sigma(E[k]) < \sigma(E)$, also when the output is empty. The interpretation is the expression's own configuration minimum; it uses no covering expression.

10. Supplementary finite checks and their limits

The accompanying Python programs independently implement the recursive operation and the finite conditions. They are supplementary reproducibility material, not a formal verification of the proof.

- A breadth-first run generated 50,001 distinct states from seeds of depths zero through four, using operation parameters zero through three and a size cutoff of 65 ordinary-plus-entry vertices. It processed 20,719 states and made 82,876 operation calls, including four on the empty expression, and checked the other 82,872 outputs for strict syntax decrease. All five admissibility conditions and the profile condition passed on the processed states. The 29,282 queued states were not included in that structural-condition count.
- The ordinary occurrence-map graph construction was compared with the directly parsed output graph on 43,996 ordinary inputs at parameters zero, one and two, giving 131,988 actual graph comparisons. The full loop visited all 50,001 generated states at those parameters (150,003 visits); its empty, deletion and special inputs were skipped.
- From a deterministic sample of 2,500 generated states, the cap check considered 350,387 candidate upward cuts. Exactly 77,604 satisfied all tested admissibility conditions (including the additional profile test used by the implementation). Every such cap strictly covered its target at parameter $\mu(B) + 1$.

These checks establish only their finite test results. They neither prove the universal finite lemmas nor check set-theoretic certificates, condensation, the minimal-counterexample argument, or the Mahlo axiom. The mathematical proof above is separate from those bounded computations.